

COLLEGE LAN NETWORK

Design & Implementation Using Cisco Packet Tracer

Network Engineering & Security Project

Project Overview

A multi-department college network designed and implemented in Cisco Packet Tracer, featuring VLAN segmentation, OSPF dynamic routing, DHCP, DNS, HTTP services, and ACL-based traffic control.

Technologies Used

Cisco Packet Tracer · IPv4 · VLAN · OSPF · DHCP · DNS · HTTP · ACL

Developed By

Aashish Karki

BCSIT Student (Cyber security enthusiast)

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1. Title

Design and Implementation of a College Network Using Cisco Packet Tracer

2. Objectives

- To design a practical college network topology using Cisco Packet Tracer.
- To configure LAN connectivity for different college departments.
- To divide the network into logical departments using VLANs.
- To configure dynamic routing using OSPF.
- To configure routers as DHCP servers for automatic IP assignment.
- To configure DNS and HTTP servers and verify access using a domain name.
- To configure an Access Control List (ACL) to restrict selected network traffic.
- To test and verify connectivity and the operation of the configured network services.

3. Theory

3.1 Switch

A switch is a network device that connects multiple devices within a LAN. It forwards Ethernet frames based on MAC addresses learned on its ports. In this project, switches are used to connect PCs and servers within each department and to assign access ports to the appropriate VLAN.

3.2 Router

A router connects different IP networks and forwards packets between them. In this project, each department has a router interface for its local network, while the routers are interconnected to form the routed backbone. The routers also provide DHCP service to their respective department networks.

3.3 LAN

A Local Area Network (LAN) connects devices within a limited geographical area. Each college department in the project has its own LAN subnet, allowing the departments to be addressed and routed separately.

3.4 VLAN

VLAN stands for Virtual Local Area Network. A VLAN logically separates devices into different broadcast domains on a switch. In this project, VLAN 10 is used for Administration, VLAN 20 for Computer Science, VLAN 30 for Management, VLAN 40 for the Computer Lab, VLAN 50 for the Library, and VLAN 60 for the Server Room.

3.5 OSPF

OSPF (Open Shortest Path First) is a dynamic link-state routing protocol. It allows routers to exchange routing information and automatically learn remote networks. OSPF Area 0 is used in this project so that the six routers can dynamically exchange the department and inter-router networks.

3.6 DHCP

DHCP (Dynamic Host Configuration Protocol) automatically provides hosts with network configuration such as IP address, subnet mask, default gateway and DNS server. In this project, the routers act as DHCP servers for the department LANs, while the PCs obtain their addresses automatically.

3.7 DNS

DNS (Domain Name System) translates human-readable domain names into IP addresses. A dedicated DNS server is configured with the domain name college.com, which resolves to the HTTP server address.

3.8 HTTP Server

HTTP (Hypertext Transfer Protocol) is used for accessing web resources. The HTTP server in the project hosts the college web page and is accessed using the domain name <http://college.com>.

3.9 Access Control List

An Access Control List (ACL) is a set of rules used to permit or deny network traffic. ACL entries are evaluated in sequence. In this project, an extended ACL is applied to the Computer Lab router to deny traffic from the Computer Lab network to the Administration network while permitting other IP traffic.

4. Network Topology

The proposed college network consists of six departments/areas: Administration, Computer Science, Management, Computer Lab, Library, and Server Room. Each area has a router and switch. The routers are interconnected in a ring topology to provide multiple paths between parts of the network. PCs are connected to departmental switches, while the HTTP and DNS servers are connected to the Server Room switch.

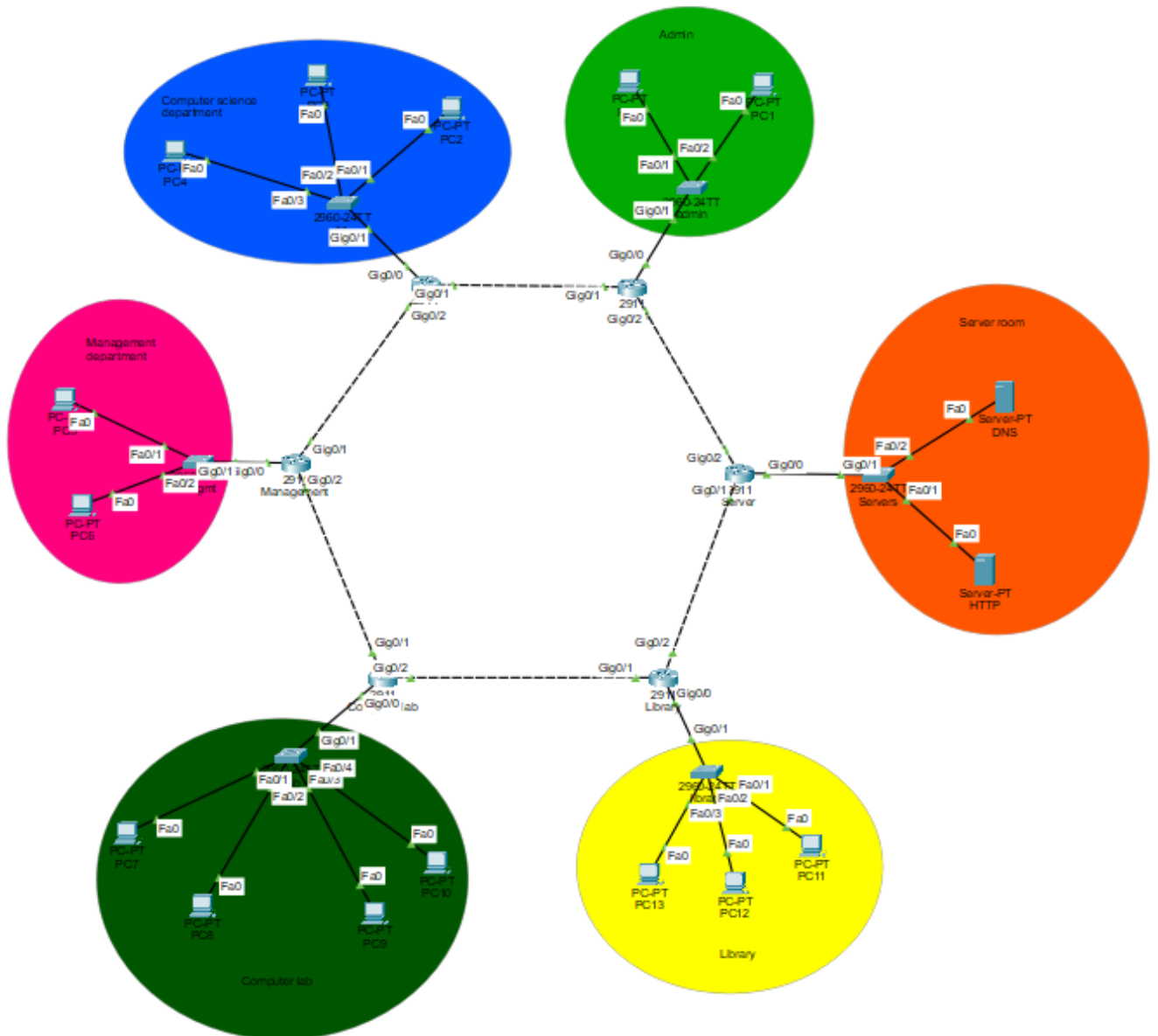


Figure 1: Network Topology of college

5. Network Design and Description

5.1 Departments and Devices

The network was implemented with six routers and six 2960 switches. The Administration department contains two PCs, Computer Science contains three PCs, Management contains two PCs, the Computer Lab contains four PCs, and the Library contains three PCs. The Server Room contains one HTTP server and one DNS server.

5.2 IP Addressing Plan

Area	VLAN	Network	Gateway
Administration	10	192.168.10.0/24	192.168.10.1
Computer Science	20	192.168.20.0/24	192.168.20.1
Management	30	192.168.30.0/24	192.168.30.1
Computer Lab	40	192.168.40.0/24	192.168.40.1
Library	50	192.168.50.0/24	192.168.50.1
Server Room	60	192.168.60.0/24	192.168.60.1

5.3 Inter-Router Networks

Router Link	Network	Assigned Addresses
Admin–CS	10.0.12.0/30	Admin: 10.0.12.1, CS: 10.0.12.2
CS–Management	10.0.23.0/30	CS: 10.0.23.1, Management: 10.0.23.2
Management–Lab	10.0.34.0/30	Management: 10.0.34.1, Lab: 10.0.34.2
Lab–Library	10.0.45.0/30	Lab: 10.0.45.1, Library: 10.0.45.2
Library–Server	10.0.56.0/30	Library: 10.0.56.1, Server: 10.0.56.2
Server–Admin	10.0.61.0/30	Server: 10.0.61.1, Admin: 10.0.61.2

6.Router Configuration

All six routers were assigned Class C private IPv4 networks for their department LANs. The inter-router links use /30 subnets to provide two usable addresses per point-to-point link. The router interfaces were enabled using the no shutdown command.

6.1 Router Interface Configuration

The router interfaces were configured with the addressing plan described in Section 5. GigabitEthernet0/0 connects each router to its departmental switch, while GigabitEthernet0/1 and GigabitEthernet0/2 are used for inter-router links.

Turn the routers interface on:

```
enable
configure terminal
interface gigabitEthernet 0/0
no shutdown
exit
interface gigabitEthernet 0/1
no shutdown
exit
interface gigabitEthernet 0/2
no shutdown
exit
end
```

Configure IP for routers:

Admin router:

```
enable
configure terminal

interface g0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 10.0.12.1 255.255.255.252
no shutdown
exit

interface g0/2
ip address 10.0.61.2 255.255.255.252
no shutdown
exit

end
```

CS router:

```
enable
configure terminal
interface g0/0
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.12.2 255.255.255.252
no shutdown
exit
interface g0/2
ip address 10.0.23.1 255.255.255.252
no shutdown
exit
end
```

Management router:

```
enable
configure terminal
interface g0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.23.2 255.255.255.252
no shutdown
exit
interface g0/2
ip address 10.0.34.1 255.255.255.252
no shutdown
exit
end
```

Computer lab router:

```
enable
configure terminal
interface g0/0
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.34.2 255.255.255.252
no shutdown
exit
```

```
interface g0/2
ip address 10.0.45.1 255.255.255.252
no shutdown
exit
end
```

Library router:

```
enable
configure terminal
interface g0/0
ip address 192.168.50.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.45.2 255.255.255.252
no shutdown
exit
interface g0/2
ip address 10.0.56.1 255.255.255.252
no shutdown
exit
end
```

Server router:

```
enable
configure terminal
interface g0/0
ip address 192.168.60.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.56.2 255.255.255.252
no shutdown
exit
interface g0/2
ip address 10.0.61.1 255.255.255.252
no shutdown
exit
end
```

```

Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.10.1    YES manual up          up
GigabitEthernet0/1       10.0.12.1       YES manual up          up
GigabitEthernet0/2       10.0.61.2       YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router#

```

```

Router>show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.20.1    YES manual up          up
GigabitEthernet0/1       10.0.12.2       YES manual up          up
GigabitEthernet0/2       10.0.23.1       YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router>

```

Figure 2:Router configuration result of router 1 and 2

6. VLAN Configuration

Six VLANs were created to represent the six college areas. Access ports connecting PCs and the departmental router were assigned to the corresponding VLAN.

VLAN ID	Name	Department/Area
10	ADMIN	Administration
20	CS	Computer Science
30	MANAGEMENT	Management
40	COMPUTER_LAB	Computer Lab
50	LIBRARY	Library
60	SERVERS	Server Room

Figure 3: Vlan configurations

```

Switch>
Switch>enable
Switch#configure terminal
      ^
% Invalid input detected at '^' marker.

Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: VLAN configuration on admin switch.

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#nme CS
      ^
% Invalid input detected at '^' marker.

Switch(config-vlan)#name CS
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-3
Switch(config-if-range)#switch mode access
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan20
      ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: VLAN configuration on CS switch.

```

Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name MANAGEMENT
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: VLAN configuration on Management dept switch.

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 40
Switch(config-vlan)#name COMPUTER_LAB
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: VLAN configuration on Computer-lab switch.

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 50'
      ^
% Invalid input detected at '^' marker.

Switch(config)#vlan 50
Switch(config-vlan)#name LIBRARY
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 50
Switch(config-if-range)#exit
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 50
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: VLAN configuration on Library switch.

```

Switch>enable
Switch#configure terminal
      ^
% Invalid input detected at '^' marker.

Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 60
Switch(config-vlan)#name SERVERS
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
      ^
% Invalid input detected at '^' marker.

Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 60
Switch(config-if-range)#exit
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 60
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: VLAN configuration on Server switch.

7. OSPF Dynamic Routing

OSPF process 1 was configured on all six routers in Area 0. Each router advertises its local departmental network and its connected /30 inter-router networks. The OSPF neighbor table was verified and showed FULL adjacencies. The routing table also showed routes marked with O, indicating routes learned dynamically through OSPF.

Admin router:

```
enable
configure terminal
router ospf 1
router-id 1.1.1.1
network 192.168.10.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.3 area 0
network 10.0.61.0 0.0.0.3 area 0
end
```

CS router

```
enable
configure terminal
router ospf 1
router-id 2.2.2.2
network 192.168.20.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.3 area 0
network 10.0.23.0 0.0.0.3 area 0
end
```

Management router:

```
enable
configure terminal
router ospf 1
router-id 3.3.3.3
network 192.168.30.0 0.0.0.255 area 0
network 10.0.23.0 0.0.0.3 area 0
network 10.0.34.0 0.0.0.3 area 0
end
```

Computer lab router:

```
enable
configure terminal
router ospf 1
router-id 4.4.4.4
network 192.168.40.0 0.0.0.255 area 0
network 10.0.34.0 0.0.0.3 area 0
network 10.0.45.0 0.0.0.3 area 0
end
```


Library router:

```
enable
configure terminal
router ospf 1
router-id 5.5.5.5
network 192.168.50.0 0.0.0.255 area 0
network 10.0.45.0 0.0.0.3 area 0
network 10.0.56.0 0.0.0.3 area 0
end
```

Server router:

```
enable
configure terminal
router ospf 1
router-id 6.6.6.6
network 192.168.60.0 0.0.0.255 area 0
network 10.0.56.0 0.0.0.3 area 0
network 10.0.61.0 0.0.0.3 area 0
end
```

```
Router>
Router>show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
4.4.4.4	1	FULL/BDR	00:00:34	10.0.34.2	GigabitEthernet0/2
2.2.2.2	1	FULL/DR	00:00:30	10.0.23.1	GigabitEthernet0/1

```
Router>
```

```
Router#
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
6.6.6.6	1	FULL/BDR	00:00:35	10.0.61.1	GigabitEthernet0/2
2.2.2.2	1	FULL/BDR	00:00:30	10.0.12.2	GigabitEthernet0/1

```
Router#
```

Figure 4: OSPF neighbour verifications from router 3 and 1

```

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O       10.0.12.0/30 [110/2] via 10.0.23.1, 00:58:42, GigabitEthernet0/1
C       10.0.23.0/30 is directly connected, GigabitEthernet0/1
L       10.0.23.2/32 is directly connected, GigabitEthernet0/1
C       10.0.34.0/30 is directly connected, GigabitEthernet0/2
L       10.0.34.1/32 is directly connected, GigabitEthernet0/2
O       10.0.45.0/30 [110/2] via 10.0.34.2, 00:54:48, GigabitEthernet0/2
O       10.0.56.0/30 [110/3] via 10.0.34.2, 00:53:08, GigabitEthernet0/2
O       10.0.61.0/30 [110/3] via 10.0.23.1, 00:52:42, GigabitEthernet0/1
O       192.168.10.0/24 [110/3] via 10.0.23.1, 00:58:42, GigabitEthernet0/1
O       192.168.20.0/24 [110/2] via 10.0.23.1, 00:58:42, GigabitEthernet0/1
       192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
--More--

```

Figure: OSPF routing table from router 3

8. DHCP Configuration

The routers were configured as DHCP servers for the department LANs. The default gateway and DNS server address were supplied through the DHCP pools. PCs were configured as DHCP clients and successfully received addresses from their respective departmental networks.

Administration: DHCP network 192.168.10.0/24; default gateway 192.168.10.1; DNS server 192.168.60.2.

Computer Science: DHCP network 192.168.20.0/24; default gateway 192.168.20.1; DNS server 192.168.60.2.

Management: DHCP network 192.168.30.0/24; default gateway 192.168.30.1; DNS server 192.168.60.2.

Computer Lab: DHCP network 192.168.40.0/24; default gateway 192.168.40.1; DNS server 192.168.60.2.

Library: DHCP network 192.168.50.0/24; default gateway 192.168.50.1; DNS server 192.168.60.2.

```

Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.10.1
Router(config)#ip dhcp pool ADMIN
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 192.168.60.2
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: DHCP configuration on admin router

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp ecluded-address 192.168.20.1
      ^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp excluded-address 192.168.20.1
Router(config)#ip dhcp pool  CS
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 192.168.60.2
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: DHCP configuration on CS router

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.30.1
Router(config)#ip dhcp pool MANAGEMENT
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 192.168.60.2
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: DHCP configuration on Management router

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.40.1
Router(config)#ip dhcp pool COMPUTER_LAB
Router(dhcp-config)#network 192.168.40.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.40.1
Router(dhcp-config)#dns-server 192.168.60.2
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: DHCP configuration on Computer lab router

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.50.1
Router(config)#ip dhcp pool LIBRARY
Router(dhcp-config)#network 192.168.50.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.50.1
Router(dhcp-config)#dns-server 192.168.60.2
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure: DHCP configuration on Library router

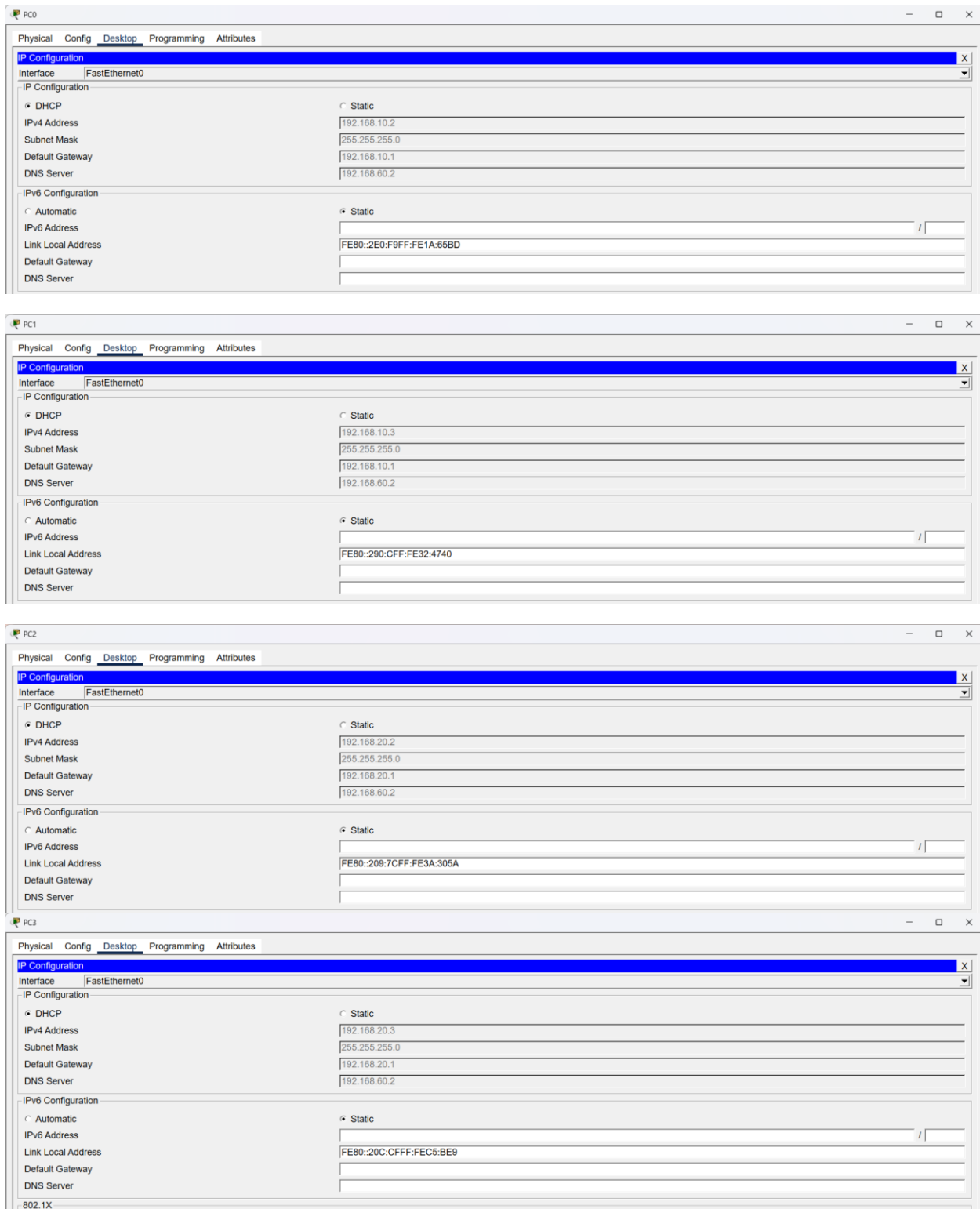


Figure : PC receiving an IP address through DHCP

9. Server Configuration

9.1 DNS Server

The DNS server was assigned a static IP address of 192.168.60.2/24 with default gateway 192.168.60.1. DNS service was enabled and a forward lookup record was created so that college.com resolves to the HTTP server at 192.168.60.3.

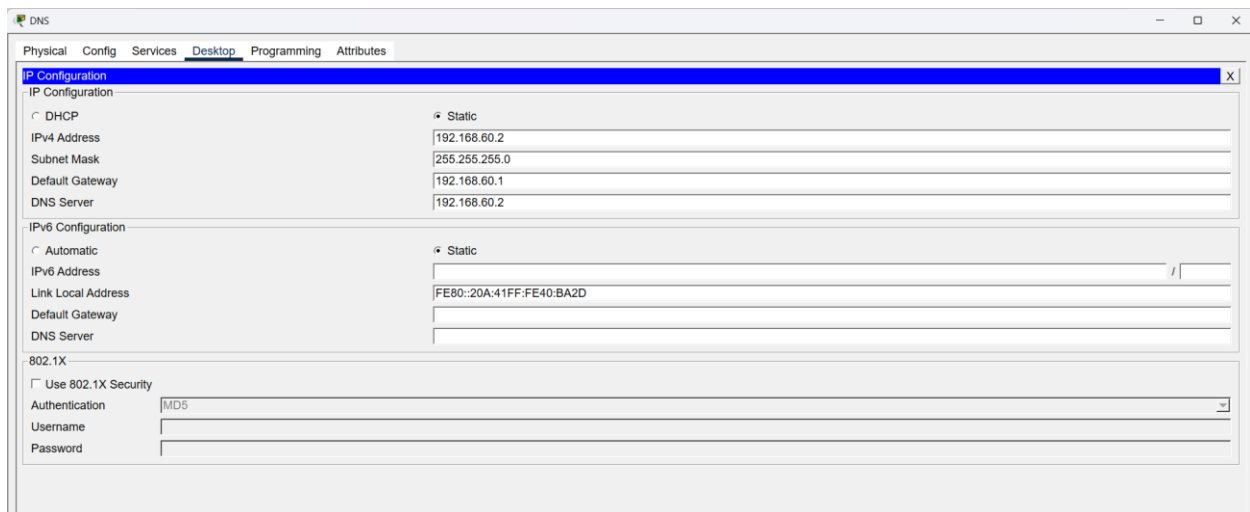


Figure 5: Manual IP configuration on DNS server

9.2 HTTP Server

The HTTP server was assigned a static IP address of 192.168.60.3/24 with default gateway 192.168.60.1 and DNS server 192.168.60.2. HTTP service was enabled. The server's default Packet Tracer web page was used to verify HTTP access.

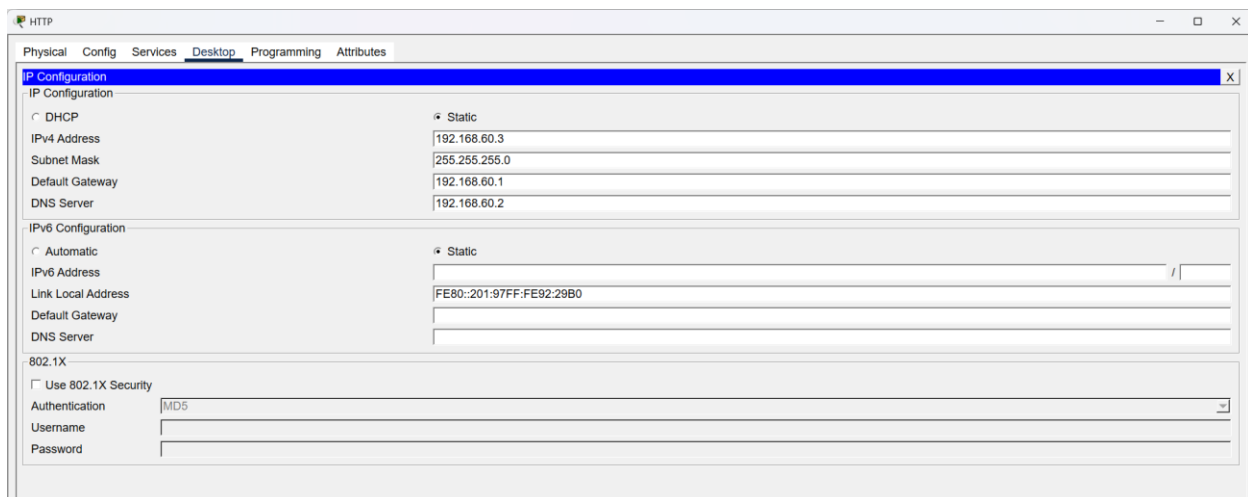


Figure 6: Manual IP configuration on HTTP server

10. Access Control List Configuration

An extended ACL numbered 100 was configured on the Computer Lab router. The ACL denies IP traffic from the Computer Lab network 192.168.40.0/24 to the Administration network 192.168.10.0/24. A permit ip any any statement was added so that other traffic remains allowed. The ACL was applied inbound on GigabitEthernet0/0, the interface connected to the Computer Lab LAN.

The ACL rule used was conceptually: deny traffic from 192.168.40.0/24 to 192.168.10.0/24, followed by permit ip any any.

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#access-list 100 deny ip
Router(config)#access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 7:ACL configuration on computer lab router

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure: ACL test from Computer Lab pc to admin

```
Router#
Router#show access-lists
Extended IP access list 100
    10 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255 (3 match(es))
    20 permit ip any any (12 match(es))
Router#
```

Figure: Verifying ACL

11. Testing and Verification

11.1 DHCP Testing

The PCs were configured to obtain IP addresses automatically. The received addresses were checked to confirm that each PC belongs to its intended departmental subnet.

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 192.168.60.2

IPv6 Configuration

Automatic Static

IPv6 Address

Link Local Address FE80::2E0:F9FF:FE1A:65BD

Default Gateway

DNS Server

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 192.168.10.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 192.168.60.2

IPv6 Configuration

Automatic Static

IPv6 Address

Link Local Address FE80::290:CFF:FE32:4740

Default Gateway

DNS Server

PC2

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 192.168.20.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 192.168.60.2

IPv6 Configuration

Automatic Static

IPv6 Address

Link Local Address FE80::209:7CFF:FE3A:305A

Default Gateway

DNS Server

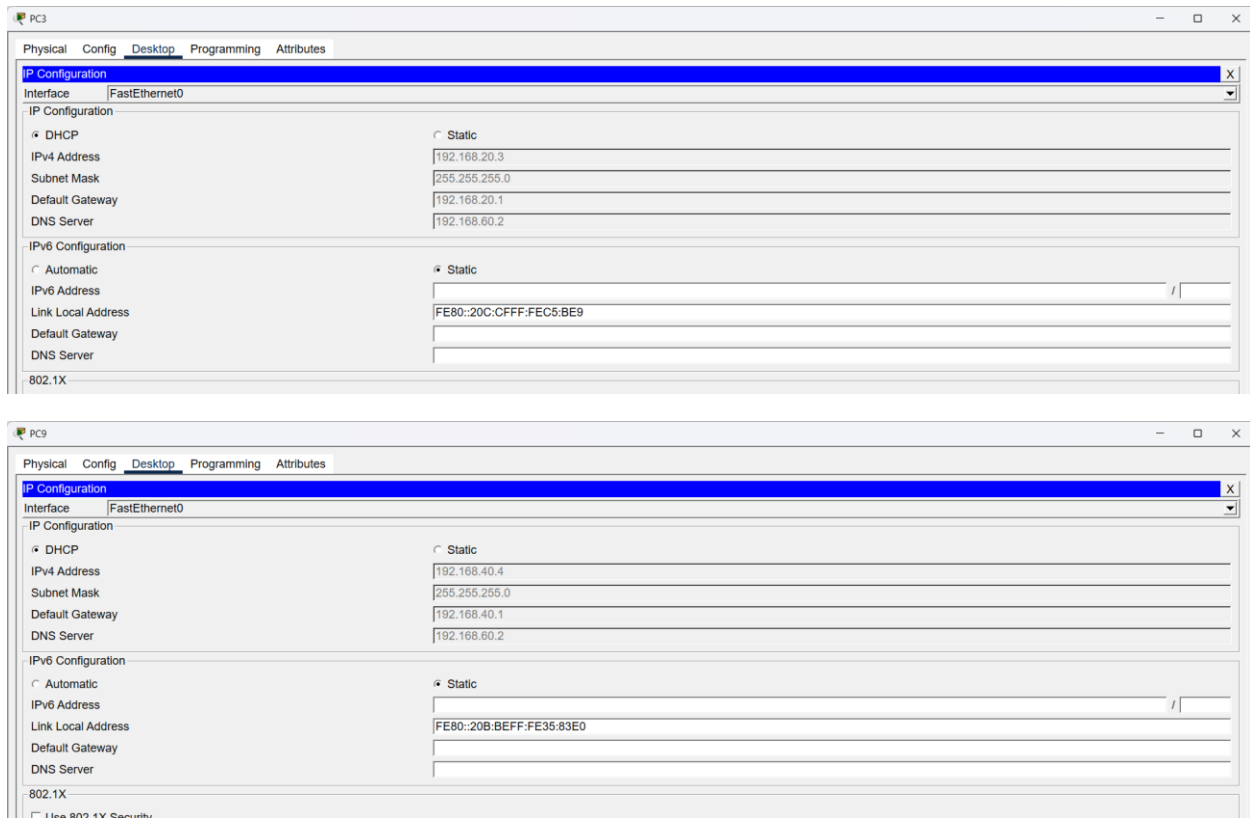


Figure 8:DHCP verification on PCs.

11.2 Connectivity Testing

Ping tests were performed between hosts and network gateways to verify LAN and routed connectivity. Successful pings to remote networks confirmed that OSPF was correctly forwarding traffic between the departmental routers.

```
C:\>
C:\>ping 192.168.50.2

Pinging 192.168.50.2 with 32 bytes of data:

Reply from 192.168.50.2: bytes=32 time<1ms TTL=125
Reply from 192.168.50.2: bytes=32 time<1ms TTL=125
Reply from 192.168.50.2: bytes=32 time=2ms TTL=125
Reply from 192.168.50.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Figure 9: Ping test between departments

11.3 DNS and HTTP Testing

The DNS and HTTP services were verified from a PC by entering `http://college.com` in the Packet Tracer web browser. The college web page was successfully displayed, confirming that DNS resolution and HTTP connectivity were operational.

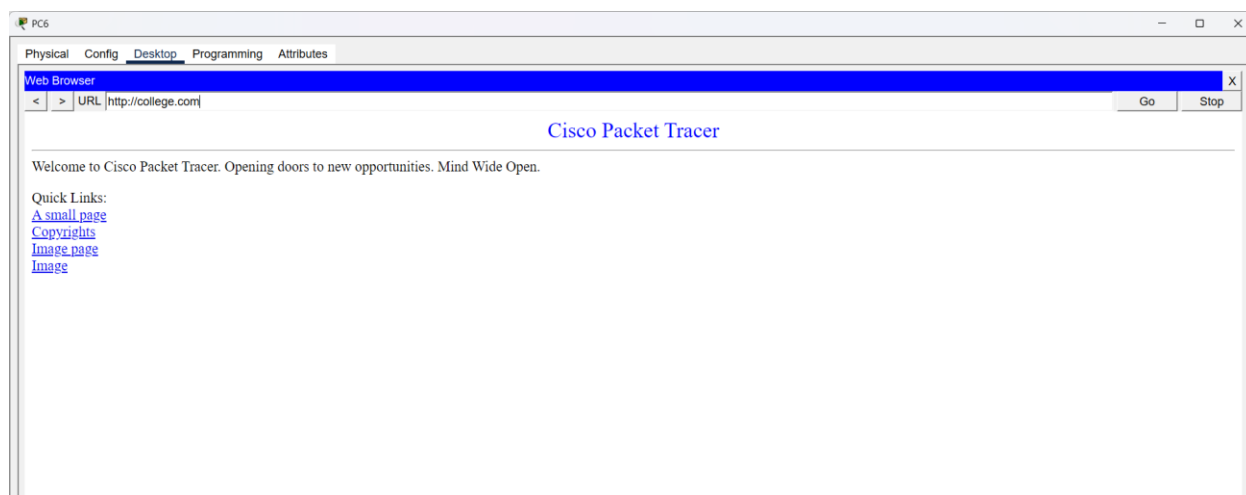
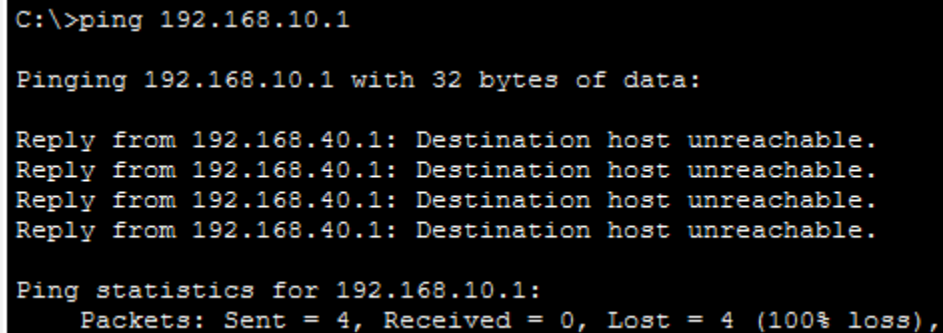


Figure 10: Accessing college.com from a PC

11.4 ACL Testing

An ACL test was performed from a Computer Lab PC. Traffic toward the Administration network was denied as intended, while traffic to permitted destinations such as the server network remained reachable.



```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 11: ACL verification and blocked traffic

12. Conclusion

The college network was successfully designed and implemented in Cisco Packet Tracer using the required networking techniques. Separate LANs and VLANs were created for Administration, Computer Science, Management, Computer Lab, Library, and the Server Room. OSPF was configured as the dynamic routing protocol and successfully formed neighbor relationships between the routers. DHCP was configured on the routers so that departmental PCs could obtain IP configuration automatically. A DNS server was configured to resolve college.com to the HTTP server, and the web service was successfully accessed using the domain name. Finally, an extended ACL was implemented to restrict Computer Lab traffic to the Administration network while allowing other required traffic. The tests performed on the topology confirmed that the designed network and its services operated as intended.